Deep Search

Deep Search I

The deep-search workflow extends the standard literature pipeline along three axes: **breadth** (multi-keyword fan-out), **depth** (full enrichment of every paper, abstract + PDF fulltext), and **archival quality** (per-paper multi-section LLM reading notes saved as standalone markdown). It is invoked from the same configuration file but a separate orchestrator script.

Configuration I

The `deep_search:` block in `manuscript/config.yaml` controls the run. The most-used knobs:

Configuration I

- `keywords` — list of free-text queries (each becomes one `SearchQuery`).
- `max_results_per_keyword` — per-keyword cap (100 by default; honoured by the aggregator after dedup).
- `sources` — backend list, same vocabulary as the standard pipeline (`arxiv`, `crossref`, `local`, `paperclip`).
- `fetch_abstracts` / `fetch_fulltext` — when both are `true`, every returned paper has its abstract fetched (arXiv export API) and (where a PDF URL is available) its fulltext extracted via `pypdf`. Cached on disk under `output/cache/abs/` and `output/cache/pdf/`.
- `llm_per_paper` — when `true` and Ollama is reachable, each paper gets a multi-section markdown reading note generated by the local LLM. When `false`, or when the LLM stack is genuinely unreachable at runtime, the per-paper note is written with only the abstract / fulltext-excerpt sections; the synthesis section is omitted entirely (no placeholder text). The composer's Supplemental S1 also drops the per-paper synthesis subsection in that case rather than emitting empty rows.
- `write_unified_bibtex` / `unified_bibtex_path` — when `true`, a deduplicated, collision-suffixed BibTeX file is written under `manuscript/references_deep.bib` so it can be cited from the manuscript via Pandoc `[@key]` syntax exactly the same way as the hand-curated `references.bib`.

Pipeline shape I

The deep-search pipeline reads `project_config.deep_search` from `manuscript/config.yaml`, runs one `SearchQuery` per keyword (capped at `max_results_per_keyword`), aggregates the per-backend results through `LiteratureClient`, enriches every paper via `AbstractFetcher` and `FulltextFetcher`, generates collision-free citation keys with `paper_to_bibentry`, and (when `llm_per_paper: true` and Ollama is reachable) calls the local LLM with `DEEP_PROMPT` to produce a seven-section reading note per paper. The per-keyword outputs are then merged by `merge_papers` to form a deduplicated aggregate roster which is written to `manuscript/references_deep.bib`, `output/deep_search/aggregate.json`, and `output/deep_search/aggregate_report.md`. A Mermaid flowchart in this subsection illustrates the same flow.

Pipeline shape I

```
flowchart TB
    CFG[deep_search block of config.yaml]
    CFG --> KW{{For each keyword}}
    KW --> SEARCH[SearchQuery ·
    max_results_per_keyword]
    SEARCH --> CLIENT[LiteratureClient<br/>aggregates
    arxiv · crossref · local · paperclip]
    CLIENT --> ENRICH[AbstractFetcher +
    FulltextFetcher<br/>cached on disk]
    ENRICH -->
    KEYS[paper_to_bibentry<br/>collision-free keys]
    KEYS --> LLMQ{LLM per-paper?}
    LLMQ -- yes --> LLM[Ollama
    LLMClient<br/>DEEP_PROMPT · 7 sections]
    LLMQ -- no --> SKIP[skip synthesis · no
    placeholder]
    LLM --> NOTE[output/deep_search/&lt;slug&gt;/per_
    paper/&lt;safe_id&gt;.md]
```

Pipeline shape II

```
SKIP --> NOTE
NOTE --> KR[KeywordResult]

KR --> AGG[merge_papers across all keywords]
AGG --> BIB[unified
BibTeX<br/>references_deep.bib]
AGG --> AJSON[aggregate.json]
AGG --> AMD[aggregate_report.md]

classDef cfg
fill:#0f172a,stroke:#0f172a,color:#fff
classDef proc
fill:#1e3a8a,stroke:#0f172a,color:#fff
classDef llm
fill:#7c2d12,stroke:#0f172a,color:#fff
classDef out
fill:#0f766e,stroke:#0f172a,color:#fff
class CFG,KW,LLMQ cfg
```

Pipeline shape III

```
class SEARCH,CLIENT,ENRICH,KEYS,KR,AGG proc
class LLM,SKIP llm
class NOTE,BIB,AJSON,AMD out
```

LLM prompt I

The deep-search prompt is richer than the standard `synthesis.PROMPT_PER_PAPER`. It produces a self-contained reading note suitable for archival without re-reading the paper:

LLM prompt

```
## Contribution
One paragraph stating the paper's central claim and
why it is novel.

## Method
2-4 bullets describing the technical approach.

## Evidence
2-4 bullets describing experiments / proofs.

## Limitations
1-3 bullets covering caveats and what the paper does
NOT address.

## Connections
1-3 bullets relating this paper to other work in the
field, citing only
papers explicitly named in the input.
```

LLM prompt I

The LLM is duck-typed as `Callable[[str], str]` so tests pass a deterministic local callable that returns real, well-formed reading-note text; runtime callers wrap an `infrastructure.llm.LLMClient` with `seed=42, temperature=0.0` for reproducibility. When the LLM stack is unreachable, callers pass `None` and the per-paper synthesis stage is skipped entirely — no placeholder text is ever written into the archive.

On-disk layout I

A successful deep-search run writes the following artefacts (paths shown relative to the project root); a Mermaid tree in this subsection depicts the same hierarchy:

On-disk layout I

- `output/deep_search/aggregate.json` — every unique paper across keywords.
- `output/deep_search/aggregate_report.md` — cross-keyword markdown summary.
- `output/deep_search/run_summary.json` — one-line metadata for the run.
- `output/deep_search/<keyword_slug>/papers.json` — enriched per-keyword paper list.
- `output/deep_search/<keyword_slug>/reading_report.md` — per-keyword markdown summary.
- `output/deep_search/<keyword_slug>/per_paper/<safe_id>.md` — one reading note per paper.
- `manuscript/references_deep.bib` — auto-generated, deduplicated unified BibTeX file.

On-disk layout II

```
flowchart TB
    P["projects/templates/template_search_project/"]
    P --> OUT["output/deep_search/"]
    P --> BIB[manuscript/references_deep.bib<br/>auto
    -generated · deduplicated]

    OUT --> AGG[aggregate.json<br/>every unique
    paper]
    OUT --> AMD[aggregate_report.md<br/>cross-keyword
    summary]
    OUT --> RS[run_summary.json<br/>one-line
    metadata]
    OUT --> KW1["&lt;keyword_slug_1&gt;/"]
    OUT --> KW2["&lt;keyword_slug_2&gt;/"]
    OUT --> KW3["&lt;keyword_slug_3&gt;/"]

    KW1 --> KW1_P[papers.json]
    KW1 --> KW1_R[reading_report.md]
```

On-disk layout III

```
KW1 --> KW1_PP["per_paper/<br/>SAFE_ID.md per
paper"]

classDef d fill:#0f172a,stroke:#0f172a,color:#fff
classDef f fill:#0f766e,stroke:#0f172a,color:#fff
classDef bib
fill:#1e3a8a,stroke:#0f172a,color:#fff
class P,OUT,KW1,KW2,KW3,KW1_PP d
class AGG,AMD,RS,KW1_P,KW1_R f
class BIB bib
```

Determinism I

Three caches make this workflow replayable:

Determinism I

1. SearchCache (`output/search/cache/search_<hash>.json`) — keyed on canonical query identity per keyword.
2. Abstract cache (`output/cache/abs/<safe_id>.txt`).
3. Fulltext cache (`output/cache/pdf/<safe_id>.{pdf,txt}`).
4. LLM seed (`project_config.deep_search.llm_seed: 42`) + temperature (`0.0`).

Determinism I

Commit any subset of these caches to version control to freeze a run.

Paperclip backend I

The Paperclip backend is opt-in. Enable it by:

Paperclip backend I

1. Creating `projects/templates/template_search_project/.env` (gitignored) with your `PAPERCLIP_API_KEY=gxl_…` (template at `.env.example`).
2. Including `paperclip` in `project_config.deep_search.sources`.

Paperclip backend I

The orchestrator scripts auto-load `.env` via the lightweight `src.dotenv` module before constructing the backend list. The `PaperclipBackend` adapter mirrors the wire protocol of the official `gxl_paperclip` Python SDK: `POST /papers` with an `X-API-Key` header and a JSON-RPC `tools/call` envelope. We support both the modern `structuredContent.papers` response shape and the older text-only `content[].text` shape (best-effort regex extraction).

Paperclip backend I

Failure-isolation: any per-backend error (including a method-rejection response the production endpoint may return during migration) is captured into `SearchResult.errors[paperclip]` and the run continues. See `output/deep_search/<keyword>/papers.json` and the aggregate `run_summary.json` for the recorded errors.

CLI I

```
# Run with everything from config.yaml (assumes
project_config.deep_search.enabled: true)
uv run python projects/templates/template_search_proj
ect/scripts/run_deep_search.py

# Force-enable without touching config.yaml
uv run python projects/templates/template_search_proj
ect/scripts/run_deep_search.py --enable

# Override keyword list at the command line
uv run python projects/templates/template_search_proj
ect/scripts/run_deep_search.py \
    --enable --keyword "convex optimization"
    --keyword "stochastic gradient descent"

# Skip LLM stage even when config enables it
uv run python projects/templates/template_search_proj
ect/scripts/run_deep_search.py --enable --no-llm
```

CLI II

```
# Bypass cache (writes still happen)
uv run python projects/templates/template_search_proj
ect/scripts/run_deep_search.py --enable --no-cache

# Local-corpus mode (offline · CI-friendly)
uv run python projects/templates/template_search_proj
ect/scripts/run_deep_search.py \
    --enable --corpus projects/templates/template_sea
    rch_project/data/corpus.json
```